PRML-Based Power Transformer Fault Diagnosis Using Dissolved Gas Analysis (DGA)

Pattern Recognition and Machine Learning Project Report

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Report based on the recorded experiments and results provided for this project.

Abstract

Dissolved Gas Analysis (DGA) is a widely used diagnostic approach for assessing the condition of oil-immersed power transformers. Different electrical and thermal stresses can generate characteristic mixtures of dissolved gases. This project formulates transformer fault-type identification as a six-class pattern-recognition problem using five gas measurements: hydrogen (H■), methane (CH■), ethane (C■H■), ethylene (C■H■), and acetylene (C■H■). The investigated classes are partial discharge (PD), low-energy discharge (D1), high-energy discharge (D2), and thermal faults T1, T2, and T3.

Five classifiers were compared: Gaussian Naive Bayes, Logistic Regression, an initial RBF Support Vector Machine (SVM), a tuned RBF SVM, and a tuned Random Forest. On a stratified 20% holdout test split, the Random Forest achieved 75.42% accuracy and a macro F1-score of 0.7382, outperforming the other evaluated models. A separate grouped five-fold validation that kept identical gas-reading vectors in the same fold achieved 77.51% accuracy and macro F1 of approximately 0.768. Permutation importance indicated that ethylene and acetylene were the strongest predictive features in these experiments.

Important qualification: these results are preliminary, dataset-specific estimates. The dataset contains repeated gas-reading vectors with conflicting class labels, and no independent transformer population was available for external validation. Therefore, the model should be treated as an academic decision-support prototype, not as a validated field diagnostic system.

1. Introduction

Power transformers are essential components of transmission and distribution networks. Internal faults can lead to service interruptions, equipment damage, and costly outages. Early identification of abnormal conditions is therefore important for condition monitoring and maintenance planning.

In oil-filled transformers, thermal and electrical stresses can decompose insulating oil and solid insulation, generating gases that dissolve in the oil. DGA measures the concentrations of these gases. Patterns across multiple gas concentrations can provide evidence about likely fault categories. However, gas patterns may overlap, and interpretation depends on the transformer, operating history, sampling, and diagnostic standard used.

Pattern Recognition and Machine Learning (PRML) methods can learn statistical relationships between gas measurements and previously assigned fault labels. This project compares conventional classifiers on a labelled dataset and examines whether a tree-ensemble model can distinguish six fault categories.

2. Objectives

• Inspect and validate a labelled DGA dataset containing five gas features and six fault classes.

• Preprocess the measurements consistently and establish a reproducible train/test evaluation.

• Implement and compare Gaussian Naive Bayes, Logistic Regression, SVM, and Random Forest classifiers.

• Tune selected models using training data only and compare their performance using accuracy and macro F1-score.

• Evaluate a grouped cross-validation setup that keeps identical gas-reading vectors together.

• Investigate which gas features contribute most to predictive performance and discuss limitations honestly.

3. Dataset Description

3.1 Features and labels

The working file is named *IEEE_STANDARD.xlsx*. It contains 589 rows before exact-duplicate removal and six columns: five numerical gas measurements and one categorical fault label. The label column in the source file is named "■■■■" and is renamed to "Fault Type" during analysis.

Feature	Gas name	Role
H■	Hydrogen	Numerical input feature
CH■	Methane	Numerical input feature

Feature	Gas name	Role
C■H■	Ethane	Numerical input feature
C■H■	Ethylene	Numerical input feature
C■H■	Acetylene	Numerical input feature
Fault Type	Categorical class label	Target

3.2 Fault categories

Label	Interpretation used in this project
PD	Partial discharge
D1	Low-energy discharge
D2	High-energy discharge
T1	Thermal fault below 300°C
T2	Thermal fault from 300°C to 700°C
T3	Thermal fault above 700°C

The working class counts before exact duplicate removal were D2: 149, T1: 111, T3: 104, D1: 91, PD: 74, and T2: 60. These are not perfectly balanced, so accuracy alone is not sufficient to judge performance; macro F1 and class-wise recall are also reported.

Dataset provenance note: the workbook was obtained through a public repository that describes a 589-sample, six-class DGA dataset and attributes its data source to an IEEE DataPort record (DOI: 10.21227/h8g0-8z59). The exact provenance and labelling of every row in the local workbook have not been independently verified. The report therefore treats the workbook as the working dataset rather than claiming independent verification of its source.

4. Exploratory Data Analysis and Preprocessing

4.1 Data quality checks

The exploratory checks found no missing values, no negative gas measurements, and no infinite values. All five gas features were numerical. The gas distributions were strongly right-skewed and included very large values compared with their medians, which is consistent with a dataset containing observations spanning a broad concentration range. Exact duplicate rows were removed, leaving 587 rows for the reported model experiments.

4.2 Repeated readings and conflicting labels

After exact-duplicate removal, 587 rows represented 582 unique five-gas feature vectors. Thus, five gas-reading vectors appeared more than once. For several repeated vectors, the same input values were assigned different fault labels. This is important because a supervised classifier receives identical inputs with contradictory targets, making the classification problem inherently ambiguous for those records.

The project did not arbitrarily delete or relabel these conflicting records. A grouped-validation analysis was also performed, placing identical feature vectors in the same fold to reduce the risk that the same gas vector appears in both a training fold and its validation fold. Grouping prevents that particular form of leakage but cannot resolve contradictory labels or replace external validation.

4.3 Scaling and split strategy

The holdout experiments used a stratified 80/20 train/test split with random_state=42. This produced 469 training samples and 118 test samples after duplicate removal. Feature scaling was used in the Logistic Regression and SVM pipelines, with scaling fitted on the training portion only. Hyperparameter tuning used training data only. The holdout test set was used for the final comparison.

Random Forest does not generally require standardisation because tree split decisions are not based on Euclidean distance or gradient scale. The dataset was not balanced by deleting majority-class examples; the tuned Random Forest used class_weight='balanced' during fitting.

5. Classification Methods

5.1 Gaussian Naive Bayes

Gaussian Naive Bayes applies Bayes' theorem and assumes that each feature follows a Gaussian distribution within each class, with conditional independence between features given the class. For a feature vector x and class C■, the classifier estimates P(C■|x) proportional to P(C■) times the product of the class-conditional feature likelihoods. It is computationally efficient, but the independence and Gaussian assumptions may not adequately represent strongly skewed and correlated gas measurements.

5.2 Logistic Regression

Multiclass Logistic Regression estimates class probabilities using a softmax function over linear scores. Feature scaling is useful because gas concentrations differ greatly in magnitude. A linear decision boundary may be insufficient when fault classes overlap in nonlinear ways.

5.3 Support Vector Machine

An SVM seeks a decision boundary with a large margin between classes. The radial basis function (RBF) kernel can represent nonlinear boundaries. The parameters C and gamma control the penalty for classification errors and the influence range of training observations, respectively. Grid search selected hyperparameters using macro F1 across five stratified folds on the training set only.

5.4 Random Forest

A Random Forest combines predictions from many decision trees trained with randomised samples and feature choices. It can model nonlinear interactions and does not require linear class boundaries. The tuned model used 100 trees, no maximum-depth restriction, minimum leaf size 2, class weighting balanced, and a fixed random seed. Grid search selected hyperparameters by macro F1 on the training data.

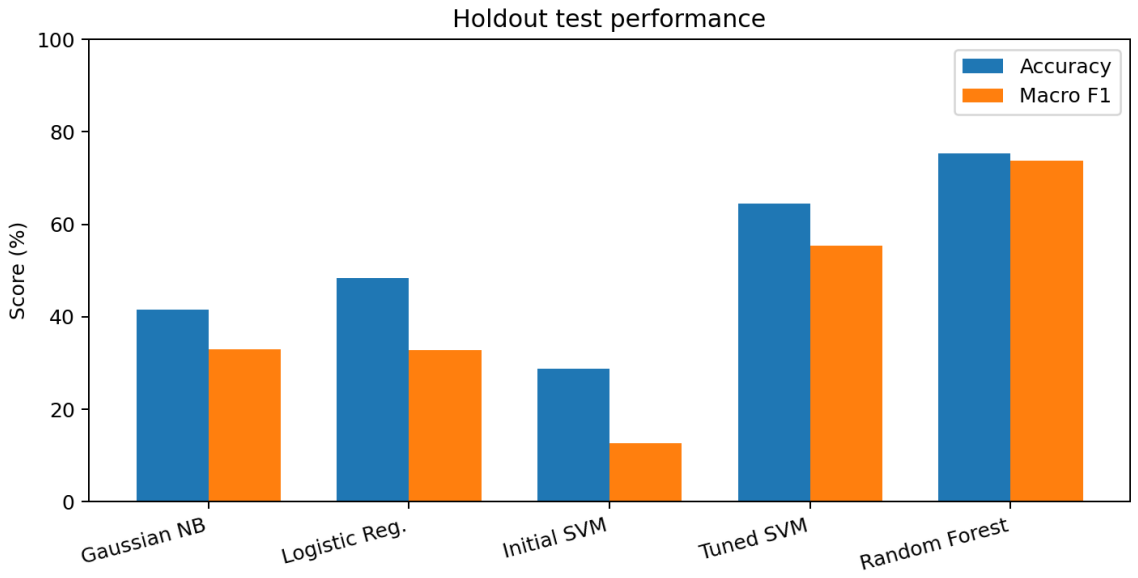
5.5 Evaluation metrics

Accuracy is the fraction of predictions that are correct. **Precision** measures how many predictions for a class are correct, while **recall** measures how many true examples of that class are detected. The **F1-score** is the harmonic mean of precision and recall. **Macro F1** computes F1 separately for each class and averages the six values equally, giving each fault class equal weight regardless of support.

6. Experimental Results

6.1 Holdout model comparison

Model	Accuracy	Macro precision	Macro recall	Macro F1
Gaussian Naive Bayes	41.53%	53.21%	40.05%	32.93%
Logistic Regression	48.31%	45.07%	39.37%	32.84%
Initial SVM	28.81%	14.85%	20.08%	12.70%
Tuned SVM	64.41%	62.46%	56.91%	55.44%
Random Forest	75.42%	74.46%	73.50%	73.82%



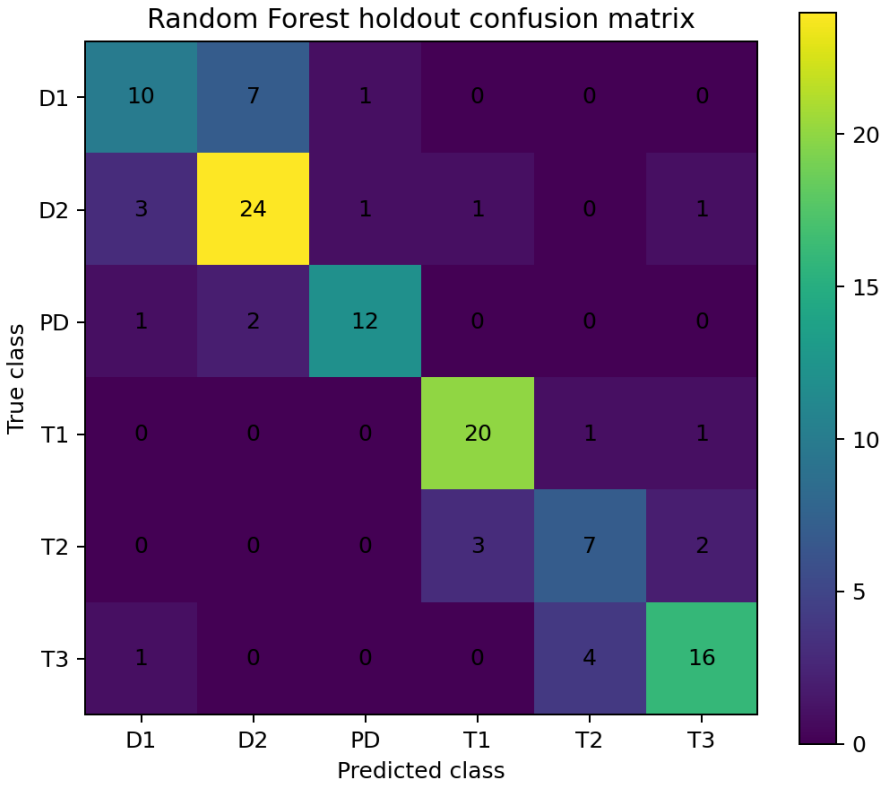
The tuned Random Forest achieved the strongest holdout result among the tested models, with 89 correct predictions out of 118. The tuned RBF SVM was the second-best model by accuracy and macro F1. The initial SVM performed poorly, while tuning produced a substantial improvement, demonstrating the importance of model configuration.

6.2 Random Forest class-wise performance

Class	Precision	Recall	F1-score	Test support
D1	0.67	0.56	0.61	18
D2	0.73	0.80	0.76	30
PD	0.86	0.80	0.83	15
T1	0.83	0.91	0.87	22
T2	0.58	0.58	0.58	12
T3	0.80	0.76	0.78	21

T1 had the highest recall (0.91), while T2 had the lowest F1-score (0.58). T2 had the smallest test support, so its estimate is relatively uncertain. D1 recall was also weaker than for most other classes. These results indicate that aggregate accuracy hides meaningful differences in class-specific reliability.

6.3 Confusion matrix



The confusion matrix shows that D1 and D2 were sometimes confused with each other, and T2 was sometimes predicted as T1 or T3. These errors are plausible for a pattern-recognition task in which gas signatures can overlap, but this experiment alone does not establish the physical cause of any specific misclassification.

6.4 Hyperparameter tuning

Model	Selection procedure	Selected configuration	Training CV result
Tuned SVM	5-fold stratified CV; macro F1	RBF; C=100; gamma=1	Macro F1 = 0.5956
Random Forest	5-fold stratified CV; macro F1	100 trees; max_depth=None; min_samples_leaf=2; class_weight=balanced	Macro F1 = 0.7482

The CV scores above are training-stage model-selection results. They are not the holdout test score and should not be interpreted as an independent final estimate. The holdout metrics in Section 6.1 were computed on the 118-sample test set.

7. Grouped Five-Fold Validation

A separate five-fold evaluation used StratifiedGroupKFold. Group identifiers were constructed from the five-gas feature vectors so that identical readings were kept in the same fold. This evaluates the model under a stricter split with respect to repeated feature vectors. The same selected Random Forest configuration was used; hyperparameters were not retuned independently inside each validation fold.

Metric	Grouped 5-fold result
Accuracy	77.51%
Macro F1 (pooled out-of-fold predictions)	0.7692
Mean fold accuracy	77.51%
Fold accuracy standard deviation	1.51 percentage points

Metric	Grouped 5-fold result
Mean fold macro F1	0.7678
Fold macro F1 standard deviation	0.0156

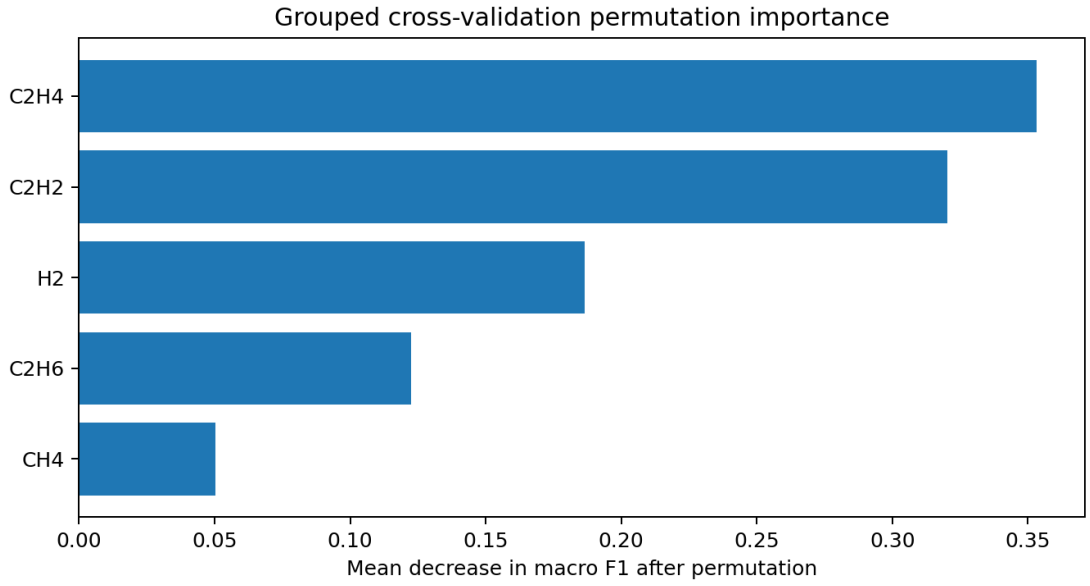
The pooled out-of-fold macro F1 (0.7692) differs slightly from the arithmetic mean of fold macro F1 (0.7678), because pooled scoring and averaging per-fold scores are different calculations. They should be reported with their definitions, not treated as interchangeable.

The grouped result is encouraging but does not prove generalisation to new transformers or other data sources. The groups are formed from identical measured feature vectors, not from transformer IDs or independent sites. If multiple samples originate from the same transformer, a future evaluation should group by transformer or site where identifiers are available.

8. Feature-Importance Analysis

Permutation importance was computed on held-out folds in the grouped cross-validation workflow. For each feature, the values indicate the average reduction in macro F1 when that feature is randomly permuted. Larger values indicate greater predictive reliance by this fitted modelling procedure; they do not establish causality or universal physical importance.

Feature	Mean macro-F1 decrease	Fold standard deviation
C■H■ (ethylene)	0.3534	0.0128
C■H■ (acetylene)	0.3202	0.0208
H■ (hydrogen)	0.1865	0.0381
C■H■ (ethane)	0.1225	0.0188
CH■ (methane)	0.0504	0.0287



C■H■ and C■H■ had the highest permutation-importance values, followed by H■. The same two features also ranked highly in the recorded tree-based feature-importance analysis. Because gas features may be correlated, permutation importance can distribute or exaggerate predictive reliance depending on the feature relationships and permutation procedure.

9. Discussion

The Random Forest provided the best overall holdout performance in the reported comparison. Its nonlinear decision boundaries and ability to model feature interactions may be useful for gas patterns that are not separable by simple linear boundaries. The tuned SVM also improved markedly over its initial configuration, showing that hyperparameter choice matters.

Performance should be interpreted cautiously. The dataset is modest in size, class frequencies differ, gas distributions are strongly skewed, and some repeated input vectors have contradictory labels. The low recall for D1 and moderate performance for T2 may be particularly important if the classifier is intended to support maintenance decisions, because different error types can have different practical consequences.

A diagnostic model should not make an operational decision based solely on a predicted label. In practice, DGA interpretation should be considered alongside transformer history, sampling quality, gas trends, other tests, applicable standards, and qualified engineering judgement.

10. Limitations and Threats to Validity

• **Dataset provenance:** the local workbook's full provenance and row-level label correctness have not been independently audited.

• **Conflicting labels:** identical gas vectors occur with different target labels, imposing a limit on learnability and complicating interpretation.

• **Small sample size:** 587 rows after exact-duplicate removal is a limited basis for a six-class diagnostic system.

• **Source and transformer dependence:** transformer identifiers, sampling context, and site-level grouping were not available in the recorded experiments.

• **Model-selection uncertainty:** hyperparameters were selected on training data, but a nested cross-validation design was not used for a fully unbiased estimate of the tuning procedure.

• **No external validation:** there is no independent dataset from another source or fleet to establish field generalisation.

• **Feature importance:** importance values describe model reliance under this dataset and are not causal explanations.

• **Clinical/engineering deployment:** no safety-critical deployment or cost-sensitive error analysis was performed.

11. Conclusion and Future Work

This project applied PRML classification methods to five dissolved-gas measurements for six transformer fault categories. Among the evaluated models, the tuned Random Forest achieved the best holdout performance, with 75.42% accuracy and a macro F1-score of 0.7382. In grouped five-fold validation that kept identical gas-reading vectors together, the Random Forest achieved 77.51% accuracy and a mean fold macro F1 of 0.7678. Permutation importance ranked ethylene (C■H■), acetylene (C■H■), and hydrogen (H■) as the most predictive features in the recorded experiment.

The results demonstrate the potential of supervised learning for exploratory DGA fault classification, but they do not yet establish a field-ready diagnostic tool. Future work should verify the original data source and labelling, obtain transformer or site identifiers, evaluate on a truly independent dataset, investigate conflicting labels with domain experts, compare additional algorithms, and report uncertainty and fault-specific error costs. Model decisions should ultimately be assessed with transformer specialists and applicable diagnostic standards.

12. Reproducibility Notes

• Working data file: IEEE_STANDARD.xlsx.

• Exact duplicate rows were removed before the reported model experiments, resulting in 587 rows.

• Holdout split: stratified 80/20, random_state=42 (469 train, 118 test).

• Tuned SVM and Random Forest hyperparameters were selected using training data only.

• Grouped validation: StratifiedGroupKFold with five folds, keeping identical five-gas vectors together.

• Saved analysis artifacts reported by the project scripts include model_comparison.csv, grouped_validation_summary.csv, permutation_importance_summary.csv, rf_fold_accuracy.png, rf_feature_importance.png, and permutation_importance.png.

• The project_results.py script records previously obtained values; it does not rerun the experiments. Preserve original scripts, workbook, dependency versions, and generated artifacts for reproducibility.

References

[1] AkhilTyagi22, “Transformer_dga,” public GitHub repository, https://github.com/AkhilTyagi22/Transformer_dga (accessed for the project dataset description).

[2] IEEE DataPort record attributed by the repository: DOI 10.21227/h8g0-8z59. The relationship between the local workbook and the cited record should be verified before final submission.

[3] Scikit-learn documentation, model selection and evaluation: https://scikit-learn.org/stable/model_selection.html

[4] Scikit-learn documentation, Random Forests: https://scikit-learn.org/stable/modules/ensemble.html#forests-of-randomized-trees

[5] IEC 60599, Mineral oil-filled electrical equipment in service — Guidance on the interpretation of dissolved and free gases analysis. Consult the edition adopted by the course or institution for formal standards-based discussion.

Before submission, check your course’s required citation style and confirm the exact dataset citation against the source file or repository.